

ADVANCED PYTHON

Applied Python Lab | 20 hours | 10 × 2-hour sessions

AUDIENCE: Students with previous programming experience, particularly in C.

Course objectives

Develop practical, idiomatic Python skills through data analysis, image processing, APIs and interactive applications.

Students learn to select libraries, experiment independently and integrate tools into small projects.

Course content

SESSION	MODULE	MAIN TOPICS
1	Python essentials	Syntax, collections, functions, strings, files and Pythonic iteration
2 - 4	Data analysis	NumPy, pandas, data cleaning and Matplotlib
5	Practical OOP	Classes, instances, attributes, methods and composition
6 - 7	Computer vision	Image manipulation, filtering and object detection with OpenCV
8	APIs and JSON	Retrieve, process and display external data
9	SciPy introduction	Interpolation, numerical routines and basic signal processing
10	Integration project	Consolidate learning in a practical mini-project

Teaching approach

Short introductions followed by guided labs in Google Colab/Jupyter, independent experimentation and progressive exercises.

> Approx. **20 minutes of concepts**, **90 minutes of practice** and **10 minutes of review per session**.

Practical work

One two-part Python **homework** (single submission), **weekly** hands-on **labs** and a final small-group **integration project**. Later activities may be adapted to student interests.

Scope: Machine learning is taught separately.

Session 9 may be replaced by an introductory to Interactive Interface (GUI) lab.